

Hyeontae Joo

Undergraduate Student

Citizenship: Republic of Korea

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Objective

Passionate about mechanical control and design, especially in applying these fields to the development of exoskeleton robots

Education

Daegu Gyeongbuk Institute of Science and Technology (DGIST), Daegu Mar. 2022 – Current

Major: Mechanical Engineering, School of Undergraduate Studies

(GPA: 3.88/4.3, Credit: 95/130)

Gosaek High school Mar. 2019 – Feb. 2022

High School Degree

Work Experience

Wearable Engineering & Assistive Robotics Lab. (Prof. Joon-Hyuk Park) Nov. 2025 – Feb. 2026

Wearable Engineering & Assistive Robotics Lab. (Prof. Joon-Hyuk Park) Jul. 2026 – Aug. 2026

Projects

Two-Segment Tendon-Driven Continuum Robot (TDCR) Nov. 2025 – Aug. 2026

- Designed a two-section TDCR with ten tendon actuators for independent bending and extension/retraction.
- Redesigned 3D-printed discs, TPU ligaments, spring backbones, and electronics housings to improve alignment, stability, and maintainability.
- Developed MATLAB simulations for workspace, tendon tension, motor torque, and collision analysis, and implemented Python-based multi-servo control.

User-Interactive Multimodal Robot Control System

Undergraduate Group Research Program (UGRP)

Mar. 2026 – Dec. 2026

- Developed a multimodal intention-recognition and control system for a 6-DOF PiPER robotic arm using visual and behavioral data from the Meta Quest Pro.

- Processed user-view, motion, and stereo camera data to estimate intended targets and desired object-placement coordinates.
- Implemented coordinate-based robot-arm control for object pick-and-place tasks and contributed to system integration and experimental documentation.

Research Interests

Humanoid Robot, Exoskeleton Robot, Robotic Arm Systems, Wearable Device Technologies for human-robot interaction and assistive applications, Continuum Robot

Skills

AutoCAD, Fusion, SolidWorks, Python, Arduino IDE, 3D Printing, MATLAB

Awards & Scholarships

DGIST Dean's List	Aug. 2026
DGIST Dean's List	Mar. 2025
Government-Funded Full-Tuition Scholarship, DGIST	Dec.2021
Granted to all DGIST students; covers the full cost of tuition for the entire program	

Professional Experience

Military Service	Apr. 2023 – Oct.2024
Republic of Korea Army, 7th Division, 5th Brigade – Fire Direction Specialist(Artillery Unit)	